



Artificial intelligence for information-driven resilience: Enhancing strategic adaptation and entrepreneurial success

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ABSTRACT

This study investigates how artificial intelligence (AI) innovation capability—the systematic integration, management, and application of AI-driven information—enhances strategic resilience and entrepreneurial success under conditions of market volatility and resource constraints. We conceptualize information-driven resilience as the organizational capacity to convert AI-enabled information flows into adaptive responses that sustain competitive functioning. Drawing on dynamic capability theory, we examine how early-stage ventures transform AI capabilities into adaptive capacity and competitive advantage. Using a multi-method approach, we collected two-wave survey data from 357 early-stage ventures in China and Europe, complemented by importance-performance map analysis (IPMA), fuzzy-set qualitative comparative analysis (fsQCA), and semi-structured interviews. Our convergent multi-method design provides complementary evidence: PLS-SEM establishes the mediation hypothesis, IPMA identifies actionable managerial priorities, fsQCA reveals equifinal configurational pathways to high performance, and interviews illuminate the underlying mechanisms. Specifically, AI innovation capability is significantly associated with strategic resilience, which is positively related to entrepreneurial success. The mediating effect of strategic resilience is robust and fully accounts for the AI capability–success relationship, indicating that the mechanism through which AI investments generate performance returns is fundamentally organizational rather than technological in nature. The value derived from AI is unlocked by first building organizational capacity for adaptive learning and agile decision-making. The study contributes to information management literature by clarifying how AI-enabled information processing capabilities enhance adaptive responses in uncertain environments, while providing practitioners with an actionable framework for developing information-driven resilience through strategic AI implementation.

1. Introduction

In the evolving landscape of information management, organizations face mounting pressure to strategically leverage artificial intelligence (AI)—the systematic integration, deployment, and application of AI technologies across organizational processes to transform data into actionable intelligence and drive adaptive responses to environmental shifts (Chau & Xu, 2025; Gama & Magistretti, 2025; Nambisan et al., 2017). This strategic AI deployment, encompassing technological integration, managerial orchestration, and innovative application, has elevated AI innovation capability to a strategic imperative for

organizational survival and competitive positioning (Ameen et al., 2024; Gupta et al., 2025; Przegalinska et al., 2025; Strauss et al., 2024). The dynamic information ecosystems of China and Europe further intensify these pressures, compelling early-stage ventures to implement advanced information processing technologies that enhance competitive positioning through superior knowledge generation and proactive responsiveness (Ku & Chen, 2024; Lengnick-Hall et al., 2011). The selection of China and Europe as our dual research contexts was deliberate and theoretically motivated. These two regions represent contrasting institutional logics that enable us to test the robustness of the AI innovation capability–strategic resilience–entrepreneurial success model across

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